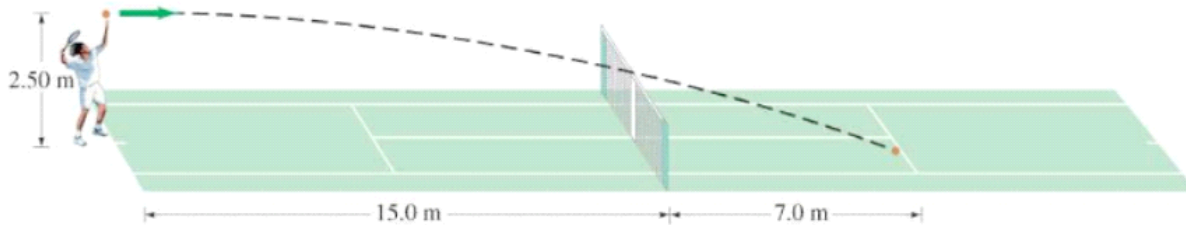


Prob 4: At serve, a tennis player aims to hit the ball horizontally as shown in figure. Ignore air resistance for all calculations.

- (a) What minimum speed is required for the ball to clear the 0.9 m high net above the 15 m from the server if the ball is launched from a height of 2.5 m?
 (b) Where will the ball land if it just clears the net?
 (c) How long will it be in air?



(a) $x = v_{0x} t$

$$y = y_0 + v_{0y} t + \frac{1}{2} a_y t^2$$

$$0.9 = 2.5 + 0 - 4.9 t^2$$

$$\Rightarrow \boxed{t = 0.57 \text{ sec}}$$

$$x = v_{0x} t$$

$$\Rightarrow v_{0x} = \frac{x}{t}$$

$$= \frac{15}{0.57}$$

$$\Rightarrow \boxed{v_{0x} = 26.3 \text{ m/s}}$$

Ans

(b) $x = v_{0x} t$

$$y = y_0 + v_{0y} t + \frac{1}{2} a_y t^2$$

$$0 = 2.5 + 0 - 4.9 t^2$$

$$\Rightarrow \boxed{t = 0.714 \text{ sec}}$$



$$x = v_{0x} t = 26.3 \times 0.714$$

$$\Rightarrow \boxed{x = 18.8 \text{ m}}$$

Ans

$$\Rightarrow \boxed{\lambda = 18.8 \text{ m}} \quad \underline{\text{Ans}}$$

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$$\boxed{t = 0.714 \text{ Sec}} \quad \underline{\text{Ans}}$$